

Cybersecurity and Zero Trust Architecture

Cybersecurity is the practice of protecting systems, networks, and data from unauthorized access and attacks. As organizations move infrastructure to the cloud, security has become increasingly important.

Traditional Security Models

Traditional security models assumed that anything inside a corporate network could be trusted. Once a user gained access to the internal network, they often had broad permissions across systems.

Zero Trust Architecture

Zero Trust is a modern security approach based on the principle of “never trust, always verify.” Every user, device, and application must continuously authenticate and authorize requests regardless of location.

Identity and Access Management

Identity and Access Management systems control who can access resources within an organization. Multi-factor authentication adds additional security by requiring multiple forms of verification.

Least Privilege Access

Least privilege means users and services should only receive the minimum permissions necessary to perform their tasks. This reduces the potential damage caused by compromised accounts.

Encryption

Encryption protects sensitive information by converting data into unreadable formats unless the correct decryption key is provided. Encryption is commonly used for both stored data and network communication.

Network Segmentation

Network segmentation divides infrastructure into isolated sections to reduce lateral movement during attacks. If one area is compromised, attackers cannot easily spread throughout the entire network.

Security Monitoring

Security teams use monitoring systems to detect suspicious behavior and potential attacks. Logs, alerts, and anomaly detection systems help organizations respond quickly to threats.

Cloud Security

Cloud environments require additional security controls including IAM policies, secure API gateways, private networking, and centralized logging. AWS provides services such as IAM, GuardDuty, and CloudTrail to improve cloud security.

Incident Response

Incident response plans define procedures for detecting, containing, and recovering from cybersecurity incidents. Effective response plans minimize downtime and data loss.

Conclusion

Modern cybersecurity focuses on continuous verification, monitoring, least privilege access, and layered defense strategies to protect cloud-native systems from increasingly sophisticated attacks.